$CO_3$: $AT_2$

**SIMATS ENGINEERING**
**ASSESSMENT TOOLS**

**COURSE NAME:** Theory of Computation **COURSE CODE:** CSA13

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**COURSE OUTCOME COVERED**

**CO3: Construct regular expressions, and use the pumping lemma and closure properti to analyze regular languages. (BL:3)**

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 5**
**Duration : 1 Hour** **Total Marks:25**

## Experiments

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***Code of Conduct:***

*I* V. Smylika (192372115) *(Name / Reg No) certify that this submission is my original work an adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

Smylika. ✓
***Signature of the student:***

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CO3: AT2

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CFG : S → 0A1, A → 0A | 1A | ε

This language generates strings that start with 0 and end with 1, with a combination of 0 and 1 in between.

Ex: 01, 001, 011, 0101

code:

```
# include <stdio.h>
# include <string.h>
int main() {
char str[100];
int n, i, valid = 1;
Printf("Enter a binary string: ");
scanf("%s", str);
n = strlen(str);
if (n<2 || str[0] != '0' || str[n-1] != '1') {
valid = 0;
} for (i=0; i<n; i++) {
if (str[i] != '0' && str[i] != '1') {
valid = 0;
break;
}}
if (valid)
Printf("string belongs to the CFG.\n");
else
Printf("string does not belong to the CFG.\n");
return 0;
}
```

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CFG: S → 0S0 | 1S1 | 0 | 1 | ε

This grammar generates palindromes over {0,1}.

Ex: 0, 1, 00, 11, 010, 101, 0110

code:

```
#include <stdio.h>
#include <string.h>
int main() {
char str[100];
int i, j, valid = 1;
printf("Enter a binary string: ");
scanf("%s", str);
i = 0;
j = strlen(str) - 1;
while (i < j) {
  if (str[i] != str[j])
    valid = 0;
    break;
}
if ((str[i] != '0' && str[i] != '1') || str[j] != '0' && str[j] !=
{
    valid = 0;
    break;
}
    i++;
    j--;
}
  if (valid)
    printf ("string belongs to the CFG.\n");
  else
    printf ("string does not belong to the CFG\n");
    return 0;
}
```

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8. $S \to 0S0 \mid A$, $A \to 1A \mid \varepsilon$

This grammar generates strings of the form:

$0^n 1^m 0^n$ where $n, m \geq 0$

Ex: 1, 111, 010, 01110, 0011100

Program:

```
#include <stdio.h>
#include <string.h>
int main(){
  char str[100];
  int n, i, left=0, right, valid=1;
  Printf("Enter a binary string:");
  scanf("%s", str);
  n= strlen(str);
  right=n-1;
  for(i=0;i<n;i++)
  { if(str[i]=='1'){
    valid=0;
  }
}
  if(valid)
    Printf("string belongs to CFG.\n");
  else
    Printf("string does not belong to the CFG \n"
  return 0;
}
```

5. CFG: $S \to 1A1A$, $A \to 0A \mid 1A \mid \varepsilon$

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```
#include <stdio.h>
#include <string.h>
int main()
{
    char str[100];
    int n, i, valid = 0;
    printf("enter a binary string: ");
    scanf("%s", str);
    n = strlen(str);
    for (i=0; i<n; i++) {
    if (str[i] != '0' && str[i] != '1') {
    Printf("string does not belong to the CFG.\n");
    return 0;
}
}
    for (i=0; i<=n-3; i++)
{
    if (str[i] == '1' &&
        str[i+1] == '0' &&
        str[i+2] == '1') {
        valid = 1;
        break;
}
}
    if (valid)
        Printf("string belongs to the CFG.\n");
    else
        printf("string does not belong to the
    return 0;
}
```

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ASSESSMENT TOOLS

UBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary (5 Marks) | Level 3 – Proficient (4 Marks) | Level 2 – Developing (2–3 Marks) | Level 1 – Beginning (0–1 Mark) |
|---|---|---|---|---|---|
| Conceptual Understanding | 30 | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor or incorrect understanding |
| Accuracy of Answer | 25 | Completely correct answer with no errors | Mostly correct with minor errors | Partially correct with noticeable errors | Incorrect or irrelevant answer |
| Application of Knowledge | 20 | Correctly applies concepts to solve the question/problem | Applies concepts with minor mistakes | Limited or partially correct application | No or incorrect application |
| Completeness of Response | 15 | Covers all required parts of the question | Covers most parts with minor omissions | Covers only some parts | Incomplete or missing response |
| Clarity & Presentation | 10 | Clear, concise, and well-organized answer | Understandable with minor issues | Somewhat unclear or poorly structured | Difficult to understand |

92/100

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